



Carbon emissions along divergent plate boundaries modulate icehouse-greenhouse climates



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The exchange of carbon between oceanic plates, the deep Earth, and the atmosphere plays a significant role in modulating global climate. Icehouse-greenhouse climate fluctuations have been attributed to changes in palaeogeography and solid Earth degassing, particularly along continental arcs, to arc weathering and to the sequestration of carbon into oceanic carbonate-rich sediments. However, the proportions of these contributions and their effect on modulating global climate are poorly constrained. Here we show that the changing balance between volcanic outgassing and carbon sequestration into oceanic lithosphere is the key driver for major climate shifts. Combined volcanic arc, mid-ocean ridge and continental rift emissions exceed carbon sequestration during greenhouse climates. In contrast, oceanic plate sequestration exceeds emissions during icehouse climates. Our results challenge the long-held view that carbon degassing along volcanic arcs is the main contributor to atmospheric carbon, suggesting instead that mid-ocean ridge and rift degassing surpass arc emissions before 100 Ma.

Earth has experienced multiple climate extremes throughout the Phanerozoic, including icehouse states in the late Ordovician, late Paleozoic and Cenozoic¹. Long-lived climate fluctuations are attributed to increased atmospheric CO₂ concentrations contributing to greenhouse climates, and lower CO₂ content contributing to icehouse climates². A key source of atmospheric carbon is from the degassing of mid-ocean ridges (MORs) and continental rifts. The rate of MOR degassing is proportional to the rate of crustal production, which has varied considerably through time but has only previously been estimated for times younger than 250 Ma³. Carbon emissions along continental rifts contribute 14.5 ± 1.5 Mt C/a at present⁴, which scales with the estimated rift length through time. Island and continental arc magmatism is widely considered to add more carbon into the atmosphere than any other geologic process^{1,5}. Volcanic arc emissions are derived from decarbonated subducting slabs⁶, and metamorphic decarbonation when arc magmas ascend through crustal carbonate platforms⁷. The metamorphic decarbonation of carbonate bodies from arc magmatism, termed “carbonate platform degassing”, has been proposed to contribute significant amounts of carbon to the atmosphere^{7–9}. At present day, ~20–30% of active volcanic outgassing (equal to 18.4 ± 4.3 Mt C/a⁶) and 80–100% of diffuse outgassing (equal to 25.4 ± 12.3 Mt C/a¹⁰) along volcanic arcs is attributed to carbonate platform degassing⁸. However, their carbon emissions through time are biased by present-day estimates because the area of extinct carbonate platforms preserved and juxtaposed with subduction zones via subduction

initiation and margin conversion has progressively increased towards the present day. From the total arc outgassing of 44.1 ± 16.6 Mt C/a, summed from active and diffusive emissions, ~13.4–44.5 Mt C/a is partitioned from carbonate platforms. The remainder is contributed by decarbonated subducting slabs and a small portion from preexisting carbon in the mantle¹¹. At the present day, the carbon emissions from arc volcanoes derived from decarbonated slabs is primarily controlled by the recycling and devolatilisation of thick calcareous deep-sea sediments down to sub-arc depths¹². However, the influx of these carbonate carbon-rich sediments into subduction zones has been significant only over the last 120 Myr¹³ due to the evolution and proliferation of planktic calcifiers that comprise deep-sea carbonates¹⁴. Prior to 120 Ma, sediment-hosted carbonate carbon subduction was insignificant¹⁴, which would reduce slab-derived carbon emissions along volcanic arcs back through time. For most studies, present-day outgassing measurements at arc volcanoes are linearly scaled back through geological time using continental arc length estimates^{1,9}, and do not account for the change in composition of the subducting slab nor in the abundance of extinct carbonate platforms along subduction zones. In this study, we track the evolution of subducting carbon reservoirs and their devolatilisation, using a plate tectonic model for the last 540 Myr¹⁵ coupled to a *Perple_X* thermodynamic model¹⁶, to more accurately reconstruct arc emissions derived from decarbonated slabs. In addition, we track the presence of carbonate platforms and their intersections with volcanic arcs to

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reconstruct the total volcanic arc emissions through the Phanerozoic. Lastly, we extend rates of MOR and continental rift degassing from 250 Ma to 540 Ma to compare their importance relative to volcanic arc degassing in icehouse–greenhouse climate variations through Earth’s history.

Results and discussion

Volcanic arc emissions from carbonate platforms

To estimate the total carbon outgassing at volcanic arcs, we first quantify carbonate platform degassing rates by reconstructing carbonate platform–subduction zone intersections over the Phanerozoic using a plate tectonic model¹⁵. Previous studies have tracked the intersections of carbonate platforms with subduction zones to estimate their relative contribution to arc emissions since 410 Ma^{17,18}. We follow a similar approach but extend the reconstruction of carbonate platforms, based on those mapped at the present day, back to 500 Ma¹⁹ and scale carbonate platform emissions relative to the rate of mantle melting, rather than scaling by arc length¹⁸. The magnitude of arc volcanism is largely controlled by the hydration of the sub-arc mantle²⁰, which in turn promotes partial melting, magmatism and associated metamorphic decarbonation when magma ascends through crustal carbonate bodies⁷. We reconstruct the oceanic plate water reservoirs

(Supplementary Fig. 1) and estimate the quantity of water leaving the subducting slab at sub-arc depths shallower than 125 km, hydrating the mantle wedge, and calculate the melting rate using a model of hydrous mantle melting²¹ (refer to Methods). The result is used to scale the present-day rate of carbonate platform degassing back through the Phanerozoic.

The carbon outflux from carbonate platforms constitutes all volcanic arc emissions before 120 Ma, gradually increasing from ~12 Mt C/a at 500 Ma to 28 Mt C/yr at the present day (Fig. 1a and Supplementary Fig. 2). This is because the surface area of carbonate platforms gradually expands through time until the present day (Fig. 2 and Supplementary Movie 1). An alternative model of crustal arc growth, based on arc magma addition rates²² scaled by total arc lengths⁵, predicts generally lower carbon emissions than our curve (Supplementary Fig. 2). The discrepancy is particularly prevalent after the mid-Cretaceous when carbon emissions drop sharply but not nearly as much as an alternate model of arc emissions⁹, which predicts much higher rates of degassing before the present day because arc lengths are scaled by zircon populations rather than plate reconstructions (Supplementary Fig. 2). While zircons may be a useful constraint on the arc length where other geological data are sparse¹, their record suffers from a preservation bias. We therefore argue that plate reconstructions could be more

Fig. 1 | Fluxes of carbon into the atmosphere and mantle.

a Contributions of mid-ocean ridge, continental rift, volcanic arc, and carbonate platform outflux to atmospheric carbon. Total volcanic arc degassing (maroon line) is partitioned into the release of carbon from subducting slabs (orange line) and degassing from carbonate platforms (green line). Mid-ocean ridge (pink line) and continental rift degassing (purple line) combine to the total outflux from divergent plate boundaries (blue line).

b Carbon is sequestered into four main volatile-carrying reservoirs in the oceanic plate: deep-sea carbonate sediments¹³ (brown), hydrothermally altered ocean crust³ (yellow line), partially serpentinised mantle lithosphere (related to plate bending⁵⁶; pink line, and slow spreading rates at MORs²⁷; grey line), and metasomatised mantle lithosphere^{24,31} (cyan line). We assume that 0–20% of deep-sea sedimentary carbon is sequestered from the atmosphere, with the remainder being sourced from shallow marine environments (refer to Methods). **c** The rate of carbon recycling into the mantle is controlled by the concentration of carbon within each reservoir, variations in the plate convergence rate and trench lengths. The increase in the subduction of sediment-hosted inorganic carbon is due to the substantial increase in deep-sea carbonates after 120 Ma¹³. Top panel indicates geological time periods (C – Cambrian, O – Ordovician, S – Silurian, D – Devonian, C – Carboniferous, P – Permian, Tr – Triassic, J – Jurassic, K – Cretaceous, Pg – Palaeogene, N – Neogene). Red – blue gradients indicate greenhouse – icehouse climates (refer to Fig. 4).

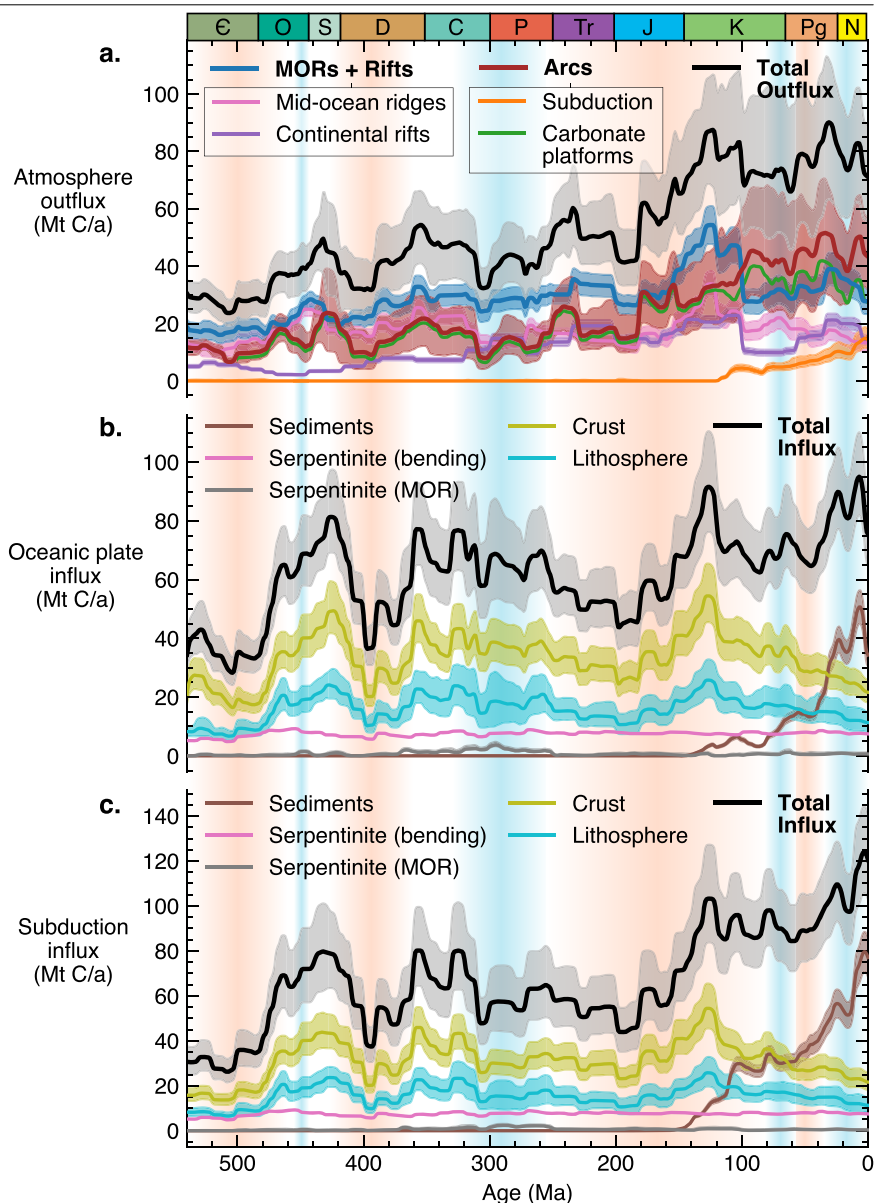
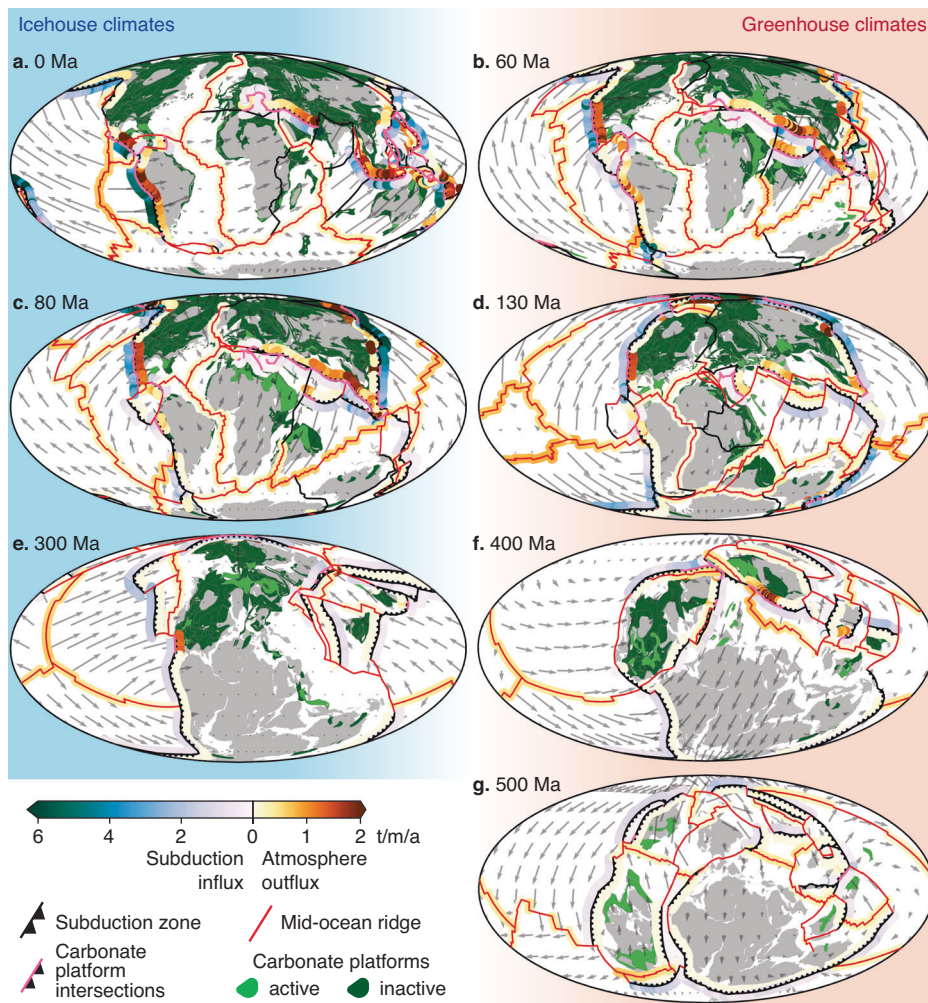


Fig. 2 | Location of carbonate platforms and carbon fluxes during key icehouse–greenhouse periods¹. Subduction influx refers to the carbon being recycled at trenches (greens–blues) in tonnes of carbon per metre along trenches per year.

Atmospheric outflux refers to carbon emitted at MORs, continental rifts, and volcanic arcs (orange colours) in tonnes of carbon per metre along each topological segment per year. Carbonate platforms are indicated by green polygons, where light green indicates active platforms, and dark green indicates older inactive platforms. Active and inactive carbonate platforms are treated the same in our model. Note that the total carbonate platform area increases cumulatively towards the present day and represents platforms that are currently preserved and have been mapped. The initial regions would have been larger prior to the platforms being partially eroded and buried. Volcanic arcs emit more carbon where subduction zones intersect carbonate platforms, highlighted by violet trench segments.

Arc melts interact with carbonate platforms on continental arcs at shallow depth, which dissolves CO₂ in the magma passing through them and contributes to additional carbon by magma degassing. Key icehouse–greenhouse climates are illustrated at **a** 0 Ma (Cenozoic icehouse), **b** 60 Ma (Early Paleogene greenhouse), **c** 80 Ma (Late Cretaceous “coolhouse”), **d** 130 Ma (Early Mesozoic–Cretaceous greenhouse), **e** 300 Ma (Late Paleozoic icehouse), **f** 400 Ma (Early Devonian greenhouse) and **g** 500 Ma (Early Paleozoic greenhouse).



useful for mapping continental arc lengths through time (Fig. 3a) because they assimilate a diverse range of geological data to constrain subduction zone and arc locations²³.

A limitation of our study is that we do not consider the thickness of carbonate platforms in our calculations. Estimating carbonate platform thickness through time is challenging owing to highly uncertain carbonate accumulation rates, which are sensitive to fluctuations in sea level and variations in sedimentation/erosion rates¹³. If carbonate platform decarbonation flux into the atmosphere scales with platform thickness, then our reconstructions of carbonate platform emissions would deviate from Fig. 1a. On the one hand, more carbon could be outgassed by continental arcs due to the subduction of thicker carbonate platforms; on the other hand, less carbon could be outgassed because we assume in our calculation that carbonate platforms that have been mapped withstand tectonic destruction throughout the Phanerozoic, and do not experience time-dependent carbon depletion. It is difficult to reconcile these two competing factors; nevertheless, our estimate of carbonate platform emissions are calibrated against present-day arc estimates^{6,10}, and roughly agrees with other reconstructions of carbonate platform emissions through time (Supplementary Fig. 2).

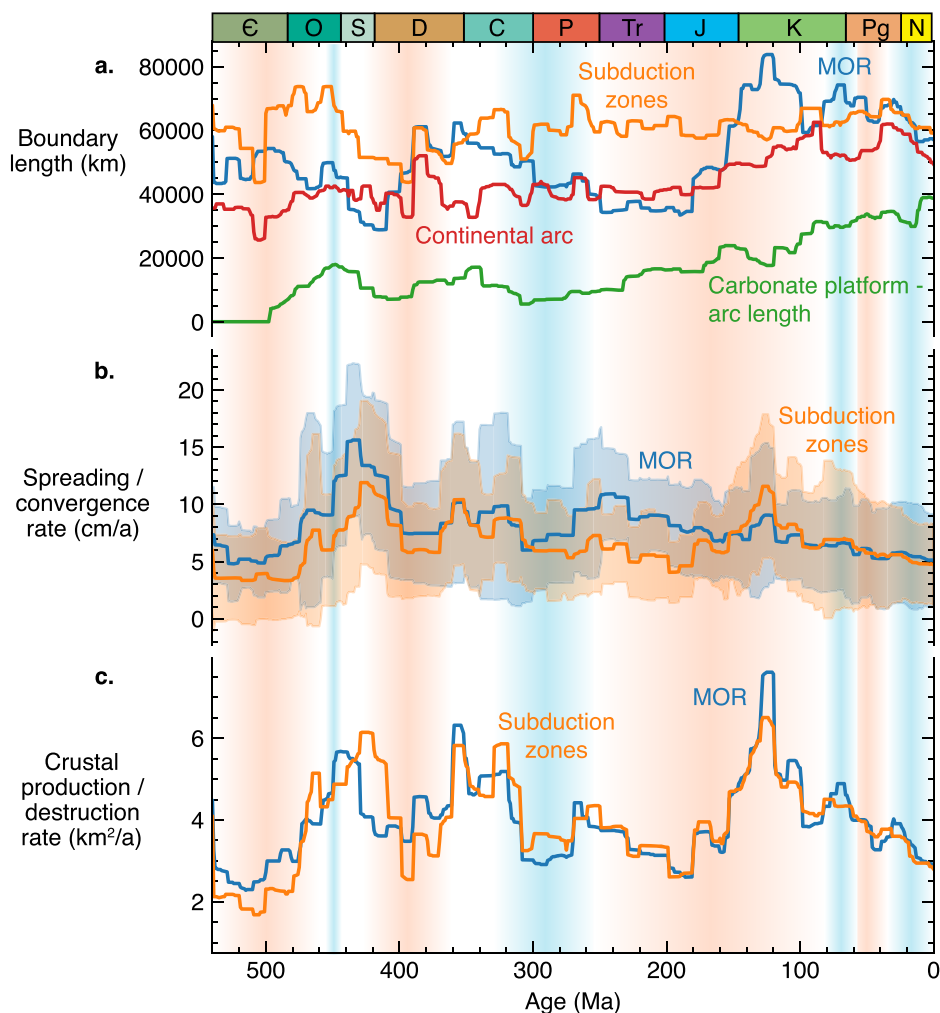
Plate influx and volcanic arc emissions from slab devolatilisation

The rate of carbon sequestration into oceanic plates is estimated using a simplified model of three major volatile-carrying reservoirs: deep-sea carbonate sediments¹³, hydrothermally altered ocean crust³, and partly serpentinised and metasomatised lithospheric mantle²⁴ (Fig. 1b). Serpentinisation related to plate bending at subduction zones stores additional carbon in the oceanic plate (4–15 Mt C/a at present day)²⁵ and is

scaled by the slab dip angle²⁶. Partially serpentinised mantle lithosphere due to slow spreading rates along MORs contributes an additional 5–11 Mt C/a into present-day oceanic plates²⁷. For deep-sea carbonate sediments, we assume that up to 20% of their total flux reflects a net increase in marine carbonate carbon sequestration since the Late Cretaceous, driven by a doubling in the flux of terrigenous sediments into the oceans during the Cenozoic²⁸, resulting in additional atmospheric carbon drawdown driven by increased continental silicate weathering (see Methods). The rate of oceanic plate influx is balanced by the subduction influx, (i.e., the amount of carbon being subducted is subtracted from the carbon being sequestered into the oceanic plates). During the subduction of carbon reservoirs stored within oceanic plates, volatiles are released from the down-going slab, stimulating partial melting and arc magmatism²¹. The carbon loss down to sub-arc depths (≤ 125 km) is calculated using a simplified thermodynamic model of each reservoir (Supplementary Fig. 3). The majority of sediment-hosted carbon is released at depths ≤ 125 km, while the other carbon-carrying reservoirs are ultimately recycled into deeper portions of the upper mantle²⁹.

Assuming that 20–30% of present-day volcanic outgassing is from carbonate platforms⁸, the remaining proportion must be derived from the decarbonation of subducting slabs¹². To estimate the outgassing related to slab outflux, we estimate the influx of carbon into major carbon-carrying reservoirs within oceanic lithosphere and track their migration as they are propelled by plate tectonics towards subduction zones (Methods, Supplementary Fig. 4). Summing all the carbon reservoirs in the oceanic lithosphere, the total carbon entering subduction zones at the present day is 101–138 Mt C/a (Fig. 1c). From our thermodynamic calculations of subducting slabs, ~59–78 Mt C/a of this total

Fig. 3 | Plate tectonic statistics through the Phanerozoic. a Lengths of mid-ocean ridges (blue), subduction zones (orange), continental arcs (red), and carbonate platforms which intersect continental arcs (green). **b** Spreading and convergence rates along MORs and subduction zones, respectively. **c** Rates of crustal production at MORs (blue) compared to crustal destruction rates at subduction zones (orange). Red – blue gradients indicate greenhouse – icehouse climates (refer to Fig. 4).



is released from the slab down to sub-arc depths, which is in broad agreement with the 46–55% of the total subducting carbon from previous carbon flux estimates²⁵. In order to fit a present-day volcanic arc degassing rate which is not attributed to carbonate platforms (11–18 Mt C/a), we estimate that ~22% of carbon released from the slab enters the atmosphere while the remainder is embedded in the overriding plate or is circulated within the upper mantle. Taking 22% of reconstructed slab carbon outflux through the Phanerozoic, we find that volcanic arc emissions are near zero prior to 120 Ma and increase to 11–18 Mt C/a towards the present day due to the subduction of deep-sea carbonate sediments³ (Fig. 1a, orange curve). A long-term increase in pelagic carbonate sediments since the mid-Cretaceous was punctuated by a doubling of carbonate carbon sequestration post-50 Ma to ~31 Mt C/a¹³. Most of this flux is thought to reflect a shift of carbonate deposition from shallow to deep water, while up to 20% of the total flux may reflect a net increase in deep-sea carbonate carbon flux since the Late Cretaceous, which reaches up to 50 Mt C/a (Fig. 1b). Consequently, present-day arc degassing rates are skewed by the relatively recent subduction of pelagic carbonate sediments (Fig. 1a). Therefore, the assumption that the present-day partitioning of subducting carbon reservoirs can be used to estimate rates of arc degassing through deep time is flawed.

Multiple studies have inferred the rate of volcanic arc degassing through deep time using the present-day rate of arc degassing paired with reconstructions of arc length from zircon populations¹ or plate reconstructions^{2,17} to explain climate variability. Our work provides an alternate calculation which accounts for the absence of significant volumes of subducting deep-sea carbonates prior to 120 Ma¹³, which are

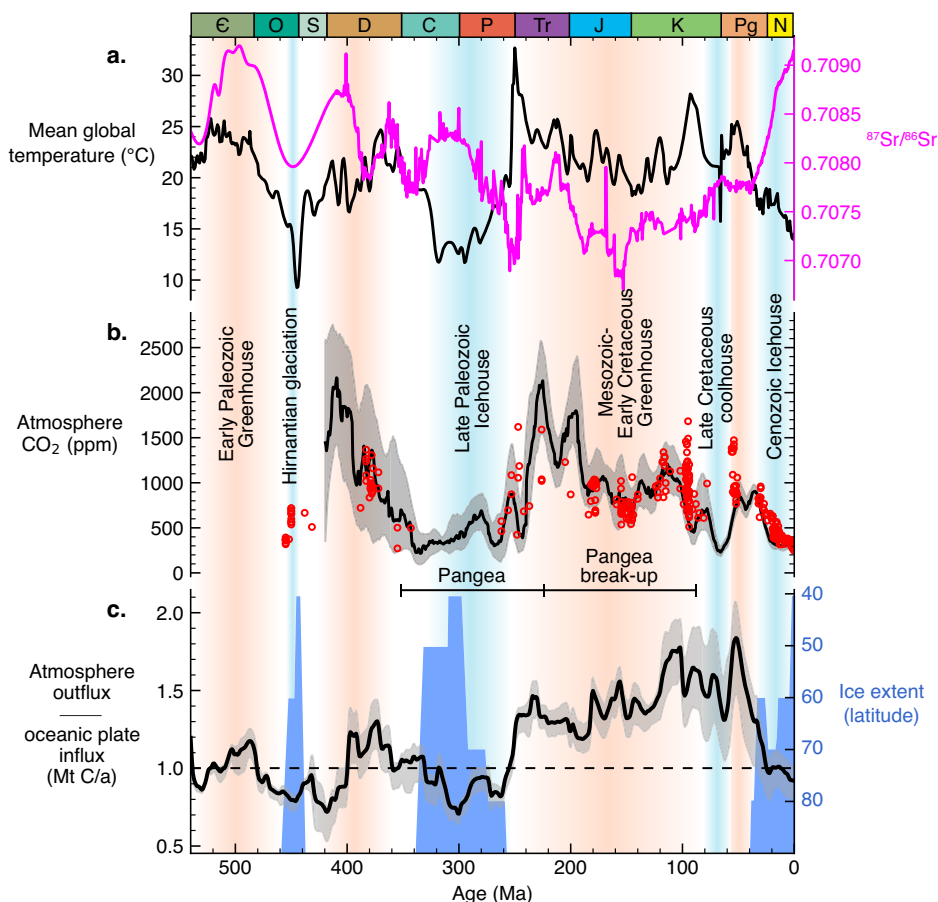
the source of most slab-derived carbon emissions from arc volcanoes (Fig. 1c). Furthermore, we incorporate plate convergence rates which are intrinsic to the amount of carbon that is recycled and outgassed along volcanic arcs (Fig. 3b). By extension, our volcanic arc degassing curve (Fig. 1a) considers the competing effects of a decline in the plate destruction rate (the product of convergence rate and trench length, Fig. 3c) and the dramatic increase in subducting carbonate sediments since ~120 Ma.

A limitation of our study is that we do not consider the deposition of continental sediments or organic carbon in our sedimentary carbon budget. However, it is estimated that deep-sea carbonate contributes over 80% of the total sedimentary carbon flux at the present day³⁰. More than 45 Mt C/a would need to be contributed by continental sediments or organic carbon into subduction zones to increase arc emissions before 120 Ma and offset emissions from MORs and continental rifts. A subduction influx of 45 Mt C/a would exceed the subduction influx from all other carbon-carrying reservoirs in subducting slabs (deep-sea sediments, altered oceanic crust, serpentinised and metasomatized mantle lithosphere) between the early Cambrian and Early Jurassic. It is therefore unlikely that the subduction of continental sediments or organic carbon could provide a sufficient boost in carbon emissions along arcs to outweigh the contribution from MOR and continental rift outgassing.

Carbon emissions along spreading ridges and continental rifts

The rate of MOR outgassing is proportional to the rate of seafloor spreading. To estimate the rate of carbon emissions at mid-ocean ridges, we connect an established model of volatile extraction³¹ to tectonic reconstructions of

Fig. 4 | Icehouse–greenhouse climate states in the Phanerozoic. Changes in the concentration of atmospheric carbon are driven by the cycling of carbon between the Earth’s mantle and surface. **a** Variations in surface temperature (black line)³⁶ and the seawater $^{87}\text{Sr}/^{86}\text{Sr}$ curve (magenta line)^{66,67}. **b** Atmospheric CO_2 concentration from a compilation of sources (black line)³⁷ overlain with CO_2 estimates from phytoplankton (red circles)⁶⁸. **c** The ratio of atmospheric outflux to oceanic plate influx (from this study) superimposed with the magnitude of glaciation events measured by the latitudinal extent of ice sheets from poles to the equator³⁸. Where the carbon outflux /influx ratio is above 1 predicts a greenhouse climate; where the ratio is below 1 predicts an icehouse climate.



MOR geometries and spreading rates through the Phanerozoic¹⁵ (Fig. 1). Our range of 10–16 Mt C/a for the present day (Fig. 1a) is consistent with previous estimates using the same volatile extraction model³. The breakup of Pangea is characterised by fast spreading rates during the Late Cretaceous (Fig. 3b) and a corresponding peak in MOR outflux of 30–40 Mt C/a (Fig. 1a). Other peaks at 360 and 450 Ma correspond to periods of increased crustal production associated with the assembly of Pangea (Supplementary Fig. 1). MOR degassing decreases from 130 Ma to the present day due to the extinction and stepwise subduction of MOR segments³. However, it is worth considering that the seafloor spreading record, and thus MOR lengths and spreading rates, are synthetic in the pre-Pangea timeframe. A combination of geological and paleomagnetic data were used as constraints on where and how fast plates diverge. However, ocean basins entirely surrounded by subduction zones carry larger uncertainties than oceans bounded by passive margins, leading to significant associated uncertainties in estimating these parameters in deep time.

The present-day rate of rift degassing is 14.5 ± 1.5 Mt C/a⁴, which we scale back through time by the rift length^{32,33}. Mapped rifts in the Cambrian and early Paleogene were formed during continental breakup and/or terrain detachment and migration, while younger rifts include a substantial portion formed in convergent settings such as back-arc basins³³. The uncertainty of estimating rift degassing is relatively high due to the preservation bias of rifts in estimating the present-day degassing rate, the age of rift initiation and cessation⁴, and because rates of rift degassing can vary along a single rift segment³⁴. The present-day rift degassing is skewed towards rifts with active plume volcanism and cratonic lithosphere involvement³⁵. Nevertheless, rift degassing combined with MOR degassing exceeds carbon emissions from volcanic arcs before 100 Ma. Prior to the middle Cretaceous, carbon emissions from MORs and rifts contributed ~62% of total carbon emissions into the atmosphere (Fig. 1a).

Icehouse–greenhouse climate variations

Compilations of palaeotemperature³⁶ and atmospheric CO_2 records³⁷ reveal the diversity of past Earth climates (Fig. 4). Continental arc weathering has been suggested as a major driver of global climate³⁸, where a high abundance of continental arc sutures within tropical latitudes leads to higher erosion rates that lower atmospheric CO_2 concentration. An increased flux of silicate-rich sediment from the continents into the oceans is correlated to higher seawater $^{87}\text{Sr}/^{86}\text{Sr}$ ratios during some greenhouse periods³⁹ (Fig. 4a); however, this is complicated by more intensive mafic weathering (associated with hydrothermal flux at MORs and off-axis alteration of the seafloor), which lowers the $^{87}\text{Sr}/^{86}\text{Sr}$ ratio⁴⁰. To more comprehensively understand past climates, the variations in $^{87}\text{Sr}/^{86}\text{Sr}$ ratios, MOR and continental arc lengths (Fig. 3a), spreading and convergence rates (Fig. 3b), and rates of oceanic crustal production and destruction (Fig. 3c) must be considered in concert with the exchange of carbon between the solid Earth and the atmosphere. By combining our estimates of carbon outgassing at volcanic arcs, MORs, and rift degassing (Fig. 1a) with carbon influx into major carbon-carrying reservoirs in oceanic lithosphere (Fig. 1b), we calculate the ratio of carbon partitioning between the Earth’s atmosphere and ocean basins (Fig. 4c, refer to Methods). Changes in this ratio predict climate trends: values >1 indicate an increase in carbon being outgassed into the atmosphere, while values <1 indicate that more carbon is being sequestered into oceanic plates. Our curve tracks most of the icehouse–greenhouse climate fluctuations through the Phanerozoic era, including the early Paleozoic greenhouse, early Devonian greenhouse, late Paleozoic icehouse, Mesozoic–Early Cretaceous greenhouse, late Cretaceous “coolhouse” climate, and the Cenozoic icehouse (Fig. 4). From a statistical analysis of mean global temperature (Fig. 4a) and the ratio of atmospheric outflux to oceanic plate influx (Fig. 4c), we find a negligible time lag between signals and a moderately strong correlation coefficient of 0.6 (Supplementary Fig. 5).

The early Paleozoic greenhouse is characterised by a high ratio of carbon outflux from MORs compared to oceanic plate sequestration (Figs. 4c, 2f), leading to moderate surface temperatures during the Cambrian (20–25 °C, Fig. 4a). Global temperatures decline from the early to middle Ordovician (20–15 °C) as the rate of crustal destruction at subduction zones exceeds crustal production at MORs (Fig. 3c). This distinct early to middle Ordovician cooling trend from ~490–430 Ma³⁶ culminated in the short-lived Hirnantian glaciation at the end of the Ordovician at 445–444 Ma⁴¹, featuring substantial ice sheets extending to 35° latitude³⁶ (Fig. 4c). The abrupt drop in global temperature has recently been attributed to an asteroid impact⁴², which cannot be reproduced by our atmospheric outflux/oceanic plate influx curve. Low seawater ⁸⁷Sr/⁸⁶Sr ratios during the late Ordovician to early Silurian have been linked to the weathering of predominantly mafic and ultramafic rocks associated with the Taconic arc-continent collision⁴³ (Fig. 4a), but may also be explained by a spike in oceanic crustal production rates from 460–435 Ma (Fig. 3c).

The high atmospheric CO₂ concentration during the early Devonian greenhouse (Figs. 2f, 4b) corresponds to a peak in our curve due to low convergence rates (3–6 cm/yr, Fig. 3b), thus higher rates of oceanic plate influx (Fig. 1b) in response to the diminished recycling of carbon-rich reservoirs into the mantle (Fig. 1c). Seawater ⁸⁷Sr/⁸⁶Sr ratios increased despite low convergence rates and a diminished potential for silicate weathering (Fig. 4a). Without a significant proportion of continental arc sutures within tropical latitudes to draw down atmospheric CO₂ via continental silicate weathering³⁸, the increase in seawater ⁸⁷Sr/⁸⁶Sr ratios likely reflects a decline in seafloor spreading rates (Fig. 3b), and thus reduced hydrothermal flux and seafloor alteration at MORs.

MOR outflux decreased from 26 to 15 Mt C/a at the onset of the late Paleozoic icehouse (Fig. 1a), due to low oceanic crustal production rates which persisted from 320–260 Ma (Fig. 3c). The combination of low MOR outflux and increased continental arc weathering³⁸, reflected in elevated seawater ⁸⁷Sr/⁸⁶Sr ratios (Fig. 4a), significantly lowered the atmospheric CO₂ content to establish one of the most extensive glaciations in geologic history⁴⁴. After a temperature spike following the late Paleozoic icehouse, likely triggered by plume-related volcanism associated with the breakup of Pangea⁴⁵ (Fig. 2d), the Mesozoic-early Cretaceous greenhouse climate was sustained by a moderately positive atmospheric outflux to oceanic plate influx ratio (Fig. 4c). A brief “coolhouse” climate during the middle Cretaceous⁴⁶, largely unexplained by other models, appears to be associated with a reduced rift degassing of 10 Mt C/a and MOR degassing of 7 Mt C/a (Figs. 1a, 2c). A return to warmer temperatures in the early Paleogene is driven by a reduced oceanic plate influx (Fig. 1b) as less carbon was being removed from the oceanic plates via subduction (Figs. 1c, 2b).

The Cenozoic icehouse is marked by a substantial increase in oceanic plate influx from 51 Mt C/a at the start of the Cenozoic (66 Ma) to 77 Mt C/a at the present day due to the proliferation of planktic calcifiers that comprise deep-sea carbonates¹⁴ (Fig. 1b). The sequestration of significant quantities of carbon into ocean basins outweighs the carbon being removed at subduction zones (Fig. 1c) and since ~22% of carbon released from the slab down to sub-arc depths is estimated to reach the atmosphere, the plate influx significantly outweighs the atmospheric outflux leading to a cooler climate. Increased continental silicate weathering, indicated by high seawater ⁸⁷Sr/⁸⁶Sr ratios (Fig. 4a), is due to the clustering of continental arcs within tropical latitudes during the Cenozoic⁴⁷ (Fig. 2a), and may have further enhanced the removal of atmospheric CO₂.

A limitation of our study is that we do not consider metamorphic CO₂ degassing along collisional orogens. The abundance of collisional zones during the Cenozoic would suggest that metamorphic CO₂ degassing should be proportionally high³⁸. While this process has been shown to emit significant quantities of carbon in the Himalayas⁴⁸ and the Tibetan Plateau⁴⁹, other studies have suggested that metamorphic CO₂ degassing is minor⁵⁰ or no more significant than other carbon sources⁵¹ at the global scale. Regardless, the drop in collisional orogens prior to the Cenozoic would significantly reduce the role of metamorphic CO₂ degassing on icehouse–greenhouse climate variations.

Conclusions

Our findings challenge the notion that continental arc emissions are the primary drivers of atmospheric carbon levels¹. Our breakdown of arc emissions from carbonate platforms and subducting plate devolatilisation reveals that this mechanism only holds for the last 100 Myr. While carbonate platform degassing emits a significant quantity of carbon at the present day and exceeds subduction outflux at volcanic arcs prior to 120 Ma, their combined contributions amount to only ~38% of total emissions, including MOR and rift degassing. Thus, carbon emissions from MORs and continental rifts are the dominant source of atmospheric carbon for most of the Phanerozoic. We find that long-term variations in global climate are primarily driven by the balance between atmospheric carbon influx (MORs, rifts and volcanic arcs), and the sequestration of carbon into oceanic plate reservoirs. The oceanic plate influx is, in turn, modulated by the recycling of carbon-rich reservoirs into the Earth’s mantle. This has significant implications for palaeoclimate reconstructions through deep time, where the balance between atmospheric outflux and oceanic plate influx is a major factor in maintaining a habitable environment for life, while large fluctuations between atmospheric outflux and oceanic plate influx may drive icehouse–greenhouse climate variations.

Methods

Reservoirs of water in the oceanic lithosphere

Oceanic mantle. Chemically-bound water is stored in the top and bottom of the oceanic mantle lithosphere. Hydration of the base of the oceanic mantle lithosphere is caused by metasomatism as new seafloor is created at mid-ocean ridges³¹,

$$F = A + B_u(u - u_{\text{ref}}) + B_d(d - d_{\text{ref}}) + B_T(T - T_{\text{ref}}) + B_c(c - c_{\text{ref}}) + B_v(v - v_{\text{ref}}) + C_v(v^2 - v_{\text{ref}}^2)$$

where F is the water influx, u is the spreading rate, d is the grainsize of mantle lithosphere, T is the ambient mantle temperature, c is the heat capacity, and v is the volatile content. Reference values for each of these parameters (denoted with _{ref}) and other constants are provided in Supplementary Table 1. The top of the oceanic mantle is affected by serpentinisation caused by mantle unroofing during the formation of new seafloor. We employ a model which relates the half-spreading rate (cm/a), u , to mantle hydration using the following function,

$$F = \frac{A}{u + B}$$

where F is the water influx, $A = 3.18 \times 10^5$ and $B = 1$. The coefficients A and B were obtained by fitting a curve to data on the mantle hydration and hydrothermal circulation at MORs³² (Supplementary Fig. 6).

Oceanic crust. The concentration of water in altered oceanic crust is partitioned into pore and chemically-bound water, which we calculate using a logarithmic model of oceanic crust age (Table 1 of ref. 53). It assumes 0.3 km of upper volcanics, 0.3 km of lower volcanics, 1.4 km of dikes and 5 km of gabbro. The bulk concentration of pore water is primarily controlled by rock type and the decay of porosity with oceanic crust age. The bulk concentration of chemically-bound water is principally controlled by rock type and density, which varies logarithmically with oceanic crust age.

Deep-sea sediments. The concentration of water in deep-sea sediments is partitioned into pore and chemically-bound water based on the thickness of sediments calculated from the distance to the nearest passive margin and age of the oceanic crust⁵⁴. From the total sediment thickness, we estimate the pore water content based on a relationship that describes

the exponential decay of porosity with depth⁵⁵,

$$z_w = \lambda \phi_s \left(1 - \exp\left(-\frac{z_{\text{sed}}}{\lambda}\right) \right)$$

where ϕ_s is the surface porosity of the sediment, λ is the decay constant, z_{sed} is the thickness of the sedimentary column, and z_w is the thickness of the water column obtained from the integrated pore space within deep-sea sediments. We multiply by the density of seawater (1,025 kg/m³) to obtain an area density of water. The remaining thickness is chemically-bound water within compacted sediments, which we assume is 7 wt. % with an average density of 1700 kg/m³. The flux of water entering subduction zones at the present day is 424 Mt/a and 131 Mt/a for pore and chemically-bound water, respectively, which is consistent with the ranges described in other studies^{56,57}.

Slab devolatilisation at sub-arc depths

Carbon and water loss at sub-arc depth (≤ 125 km) is calculated using a simplified thermodynamic model of three major volatile-carrying reservoirs in subducting slabs: deep-sea carbonate sediments¹³, hydrothermally altered ocean crust³, and partly serpentinised and metasomatised lithospheric mantle²⁴. We employ a thermodynamic lookup table approach to calculate the release of water and carbon from each reservoir in the subducting slab as a function of pressure and temperature (Supplementary Fig. 4). The bulk composition and solution models used to compute these lookup tables from Perple_X version 6.9¹⁶ are identical to those published in a previous study³. The model utilises an internally consistent thermodynamic database⁵⁸ on a grid of 313×313 nodes corresponding to a resolution of 287 bars and 3.5 K. Devolatilisation is computed by subtracting the initial H₂O and CO₂ wt% at the trench from that calculated at a depth of 125 km, which is taken as a representative average of the sub-arc depth in the mantle corresponding to the maximum generation of arc magmas⁵⁹. This number represents the total reduction in slab water and carbon concentration from the trench to sub-arc depths (≤ 125 km). We only include chemically-bound water in our calculation, since pore water would be expelled from the slab long before the slab has a chance to reach arc depths. The P-T conditions of the down-going plate at sub-arc depths (≤ 125 km) is calculated from the age of oceanic crust and dip angle of the slab at the trench. We approximate slab temperatures at the plate surface, T_s , crust-sediment interface, T_c , and the oceanic Moho, T_m , using three logarithmic fitting parameters,

$$T_s = 1331 - 58.6 \ln \Phi$$

$$T_c = 1303 - 60.23 \ln \Phi$$

$$T_m = 1622 - 132.5 \ln \Phi$$

where the temperature of each interface at 125 km depth is related to the thermal parameter, Φ , which is the product of the oceanic seafloor age, A_{plate} , orthogonal convergence velocity, v_c , and the slab dip angle²⁹, θ_{dip} ,

$$\Phi = A_{\text{plate}} v_c \theta_{\text{dip}}$$

Slab dip is calculated using a machine learning approach based on the kinematic and rheological properties of the subducting plate²⁶. The oceanic seafloor age and convergence velocity are obtained from the plate model¹⁵ using GPlately⁶⁰. The thermal parameter provides a first-order proxy for slab devolatilization²⁹: if the subducting slab is laden with volatiles, then a high Φ would lead to its efficient devolatilization.

The slab outflux, q , is obtained by (1) subtracting the initial bulk H₂O and CO₂ wt% entering the trench from that calculated at 125 km, (2) multiplying by the slab density, ρ_{slab} , and convergence rate, v_c , then (3)

integrating along the length of all subduction zones, L ,

$$q_{\text{H}_2\text{O}} = \int_{i=0}^L \left[X_i^{\text{H}_2\text{O}}(P, T) - X_i^{\text{H}_2\text{O}} \right] \rho_{\text{slab}_i} v_{c_i} dL$$

$$q_{\text{CO}_2} = \int_{i=0}^L \left[X_i^{\text{CO}_2}(P, T) - X_i^{\text{CO}_2} \right] \rho_{\text{slab}_i} v_{c_i} dL$$

This routine is repeated for each volatile-carrying reservoir in the slab (deep-sea sediments, altered oceanic crust, and serpentinized/metasomatized mantle lithosphere), which are then summed together to estimate the total slab outflux.

Volatile uptake from plate bending-related serpentinisation at subduction zones

Serpentinisation related to plate bending at subduction zones stores additional carbon and water in the down-going plate^{25,56}, which we scale by the slab dip angle²⁶. Serpentinisation within bending-induced faults occurs within the brittle part of the slab, which we take as slab temperatures below the 600 °C isotherm computed from a plate model of lithospheric cooling⁶¹. Assuming ~10% serpentinite within the brittle layer, we obtain ranges of volatile uptake in the outer rise that is consistent with previous estimates (~7.5 Mt C/a²⁵ and ~1000 Mt H₂O/a⁵⁶). A correction is also applied to the sediments based on whether a subduction zone is accreting or eroding sediment, which is proportional to the orthogonal convergence rate⁶². No sediment is accreted for convergence velocities >6.8 cm/a, and up to 40% of sedimentary carbon is accreted to the overriding plate for convergence velocities approaching 0 cm/a. For our calculations, we apply a linear gradient between these two endmembers.

Calculation of melting rates to constrain carbonate platform degassing

The quantity of water that is devolatilised from subducting plates at sub-arc depths (≤ 125 km) determines the concentration of water in the upper mantle, which controls the melting rate and magnitude of arc volcanism. We estimate the amount of water lost from the slab at sub-arc depths from thermodynamic lookup tables of three major water-carrying reservoirs (explained above). The water outflux from the slab is converted to a bulk concentration assuming an upper mantle density of 3300 kg/m³ at a distance inbound from the trench corresponding to a depth of 125 km and the dip angle of the subducting slab²⁶. The wet melt fraction is calculated using an established model of hydrous mantle melting²¹, averaged across a pressure range of 2.5–4.5 GPa at 0.5 GPa increments and assuming a mantle temperature of 1350 °C. The melting rate is calculated by multiplying the melt fraction by the density of the upper mantle, the arc depth, the convergence velocity, and integrating along all subduction zones. For the present day, we obtain a present-day melting rate of 128 Mt/a for all subduction zones and 84 Mt/a for subduction zones which intersect carbonate platforms. The latter is used to scale the rate of carbonate platform degassing through the Phanerozoic. This is dimensionalised to the present-day rate of carbonate platform degassing (13.4–44.5 Mt C/a). A similar relationship is observed if we scale the present-day carbonate platform degassing rate by the intersecting arc length (Supplementary Fig. 2). A departure between the outflux constrained by melting rates and arc lengths is particularly pronounced between 80–90 Ma due to fast convergence rates and increased melt production along western North America and eastern Eurasia, which favours increased carbonate platform degassing. Both curves assume that carbonate platforms withstand tectonic destruction and do not experience time-dependent depletion.

Calculation of the contribution of pelagic carbonate sediment accumulation to plate influx

If the enormous increase in the pelagic carbonate carbon reservoir since the Late Cretaceous¹³ merely reflected a shift from shallow to deep water carbonate deposition⁶³, then the growth of this reservoir would not have

resulted in any net increase in carbon sequestration. However, it is more likely that a portion of this increase did represent a net increase¹³, which has been substantiated by a global erosion model²⁸ demonstrating an enormous increase in continental erosion, and thus associated carbon drawdown from the atmosphere in the Cenozoic, which would have resulted in a net increase in marine carbonate carbon sequestration. We apply an empirical estimate for this carbon sink, using the following rationale: a eustatic sea-level fall after the Late Cretaceous sea-level highstand⁶⁴ would have enlarged the continental area available for continental silicate weathering. In addition, an increase in net accumulation rates of pelagic carbonate carbon starting in the Paleogene may have been driven by the emergence of major new orogens⁶⁵ and associated river systems¹³. Both effects were captured in the global erosion model for the last 100 million years²⁸, which modelled a doubling of terrigenous sediment flux to the oceans in the Cenozoic. Here, we assume that the portion of the pelagic carbonate carbon flux since 90 Ma that reflects a net increase (as opposed to a shift of carbonate sedimentation from shallow to deep water) gradually rose from 0% at 90 Ma to 20% of the total pelagic carbonate carbon flux after 50 Ma. The initial increase from 90 to 50 Ma would have been mainly driven by a global regression, while the subsequent increase was driven by the combined effect of an increased area of orogens⁶⁵ and a continuing global regression and eustatic sea-level fall⁶⁴.

Calculation of the ratio between the atmospheric carbon outflux and the oceanic plate influx

To calculate this ratio, the total carbon outgassing at mid-ocean ridges, continental rifts, and volcanic arcs (derived from slab devolatilization and carbonate platform degassing) is divided by the rate of plate influx, which is contained within the upper portion of the oceanic plates. We include all carbon stored within deep-sea sediments¹³, hydrothermally altered oceanic crust⁵, and serpentinite due to slow spreading along MORs²⁷ and plate bending at subduction zones oceanic²⁵. We exclude carbon stored within the base of the oceanic mantle lithosphere in this calculation because this carbon rarely interacts with the Earth's surface and plays no significant role in modulating greenhouse—icehouse climates.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The datasets generated and analysed in this study are available in the Zenodo repository, <https://doi.org/10.5281/zenodo.15315706>.

Code availability

Plate reconstruction software, GPlates, is freely available from www.gplates.org/download.html, and GPlately is freely available from www.github.com/GPlates/gplately. Thermodynamic modelling software, Perple_X, and associated thermodynamic databases are freely available from www.perple.ethz.ch. Python workflows we developed to estimate emissions from carbonate platforms are available from www.github.com/brmather/carbonate-platform-degassing.

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Author contributions

B.R.M. interrogated the plate reconstruction models and coupled them with the thermodynamic modelling framework, wrote the paper and prepared the figures. R.D.M. and A.D. conceived and directed the project. S.Z. digitised the carbonate platforms back to 500 Ma. All authors participated in the

discussion and interpretation of the results, as well as in the preparation of the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

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